Answer key: Lesson 7, Make Your Own Graph

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the tray. Count and write how many of each kind.

Answer: 4 shells; 5 buttons; 3 spoons

Count one kind at a time and cross out each thing. 4 shells, 5 buttons and 3 spoons. 4 + 5 + 3 = 12, the same as the things on the tray, so no thing is missed.

Watch for: Counting a thing twice.; Missing a thing in the middle of the tray.

S2. The tally shows how many things Ali counted.

Answer: 6 pens; 8 rubbers

The first group is 5. Pens: 5 and 1 more is 6. Rubbers: 5 and 3 more is 8.

Watch for: Counting a group of 5 as 4.; Counting each group as 1.

S3. Ali ate 4 apples, 6 pears and 3 plums. Draw one picture in a box for each fruit.

Answer: Apples 4 pictures; pears 6 pictures; plums 3 pictures

Draw 4 pictures in the apples row, 6 in the pears row and 3 in the plums row. Some boxes are left empty. The three rows have 4 + 6 + 3 = 13 pictures, one for each fruit.

Watch for: Filling every box in a row.; Drawing the pictures in different sizes.

S4. Look at the graph. Each picture stands for 1 pet.

Answer: 3 more cats than fish; 17 pets in all

Cats 7 and fish 4. 7 take away 4 is 3 more cats. All three rows: 7 + 4 + 6 = 17 pets.

Watch for: Adding 7 and 4 to find how many more.; Leaving out a row when finding the total.

S5. Look at the graph. Each picture stands for 1 child.

Answer: Chess; 13 children

Chess has 3 pictures. That is the shortest row, so the fewest children play chess. Ludo 5 and cards 8: 5 + 8 = 13 children.

Watch for: Choosing the longest row.; Adding the chess row as well.

S6. Look at the graph. Each picture stands for 1 stamp.

Answer: True, False, True, True

Ali 5, Mei 5 and Ben 2. Ali and Mei have the same number: true. Ben has the fewest, not the most: false. 5 take away 2 is 3, so Mei has 3 more than Ben: true. 5 + 5 + 2 = 12: true.

Watch for: Saying Ben has the most because his name is last.; Forgetting to add all three rows.

Think harder (page 3)

T1. The tally shows the things in a box. Fill in the table.

Answer: 9, 5, 6; 20

Hats: 5 and 4 more is 9. Bags: 5. Shoes: 5 and 1 more is 6. 9 + 5 + 6 = 20 things in all.

Watch for: Counting a group of 5 as 4.; Adding only two of the kinds.

T2. Study the graph. Each picture stands for 1 thing. Which question can the graph answer? Write its number in the brackets.

Answer: (2)

The graph shows how many of each kind are in the box. Question (2) asks about the number of shells and stones: 6 take away 2 is 4 more shells. The other questions are about likes, places or looks, and the graph does not show them.

Watch for: Choosing (1) because it sounds like a graph question.; Choosing a question about the prettiest.

T3. There are 16 books in all. Draw the art books in the empty boxes.

Answer: 11 maths and story books; 5 art books

Maths 4 and story 7 make 11. 16 take away 11 is 5. Draw 5 pictures in the art row. Check: 4 + 7 + 5 = 16.

Watch for: Writing 16 or 7 for the art books.; Drawing a picture in every empty box.

T4. A child in another class made a graph from the tally. One row is wrong.

Answer: Vans; 3

Cars 6 matches 6. Buses 7 matches 7. Vans have 3 strokes in the tally but 4 pictures in the graph. The vans row is wrong and should have 3 pictures.

Watch for: Choosing the longest row.; Counting the tally group as 4.

Go further (page 4)

G1. Mei's graph uses a key. Each picture stands for 2 sweets. Draw the pictures in the empty boxes.

Answer: 3 pictures; 2 pictures; 7 gums

6 toffees: 2, 4, 6 is 3 pictures. 4 lollies: 2, 4 is 2 pictures. The gums row has 3 whole pictures, 6 sweets, and half a picture, 1 more: 7 gums.

Watch for: Drawing 6 pictures for 6 toffees.; Counting half a picture as 2 or as 0.

G2. Four children have 21 stickers in all. Ben has 4 stickers. Lily has 3 more than Ben. Jamie has 2 fewer than Lily. Ravi has the rest.

Answer: Lily 7; Jamie 5; Ravi 5

Lily: 4 and 3 more is 7. Jamie: 7 take away 2 is 5. Ben, Lily and Jamie have 4 + 7 + 5 = 16. 21 take away 16 is 5, so Ravi has 5. Check: 4 + 7 + 5 + 5 = 21.

Watch for: Writing 3 for Lily (the number more, not the total).; Forgetting to take Ben away when finding the rest.

G3. Study the graph. Write a question that the graph can answer. Then write the answer. Many answers are right.

Answer: Many answers are right

Any question about the numbers on the graph is good, with the right answer. For example: How many more tulips than daisies? 6 more. How many flowers are there in all? 17. Which flowers have the fewest? Daisies. A question about which flower a child likes cannot be answered.

Watch for: Writing a question the graph cannot answer, such as which flower is nicest.; Writing a question with no answer.

Investigation (page 5): Sort, draw and ask

Many answers are right. A good graph has a title, the kinds named, one picture for each thing, all the pictures the same size, and a key. The numbers on the graph match the piles.

Look for: A child who sorts by one feature, draws one picture for each thing and writes a question the graph can answer (how many, how many more, which has the fewest). Credit a clear tally. A child who writes a question about likes has not yet met the idea: ask what the graph shows. A child who draws a graph with a key of 2 has gone further.

Exit ticket (page 5)

E1. Look at the tray.

Answer: 4 pencils; 3 rubbers

Cross out each thing as you count it: 4 pencils and 3 rubbers. 4 + 3 = 7, the same as the things on the tray.

Watch for: Counting a thing twice.; Missing a thing.

E2. Look at the graph. Each picture stands for 1 biscuit.

Answer: 5 more biscuits

Ali 7 and Mei 2. 7 take away 2 is 5 more biscuits.

Watch for: Adding 7 and 2.; Writing 7.

E3. The graph shows the biscuits eaten. Which question can the graph answer? Write its number in the brackets.

Answer: (1)

The graph shows how many biscuits each child ate, so question (1) can be answered: Siti ate 4. The graph does not say which biscuit tastes best.

Watch for: Choosing (2): the graph does not show likes.

Sneak-peek puzzles (not in the workbook)

1. 4 spoons.
2. Any clear picture: for example a row of 4 spoons, a row of 3 buttons and a row of 2 shells, one small picture for each thing.